MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE KDQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

1.2 Classification This specification covers one class of modules

Class I as specified in MIL-STD-1389.

2 APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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3 3 Failure rate requirements All modules acquired to this specification shall have a predicted failure rate no greater than 0.49 failures per million operating hours when predicted in accordance with MIL-STD-1389

3 4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDQ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3 5 5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A.

3 5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5 8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/60-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier)

3.6 Circuit pin configuration The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance The quality assurance provisions of MIL M 28787 shall apply

4.2 Acceptance tests Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified

4 4 Test procedures The module shall be tested as specified in table I

4.4 1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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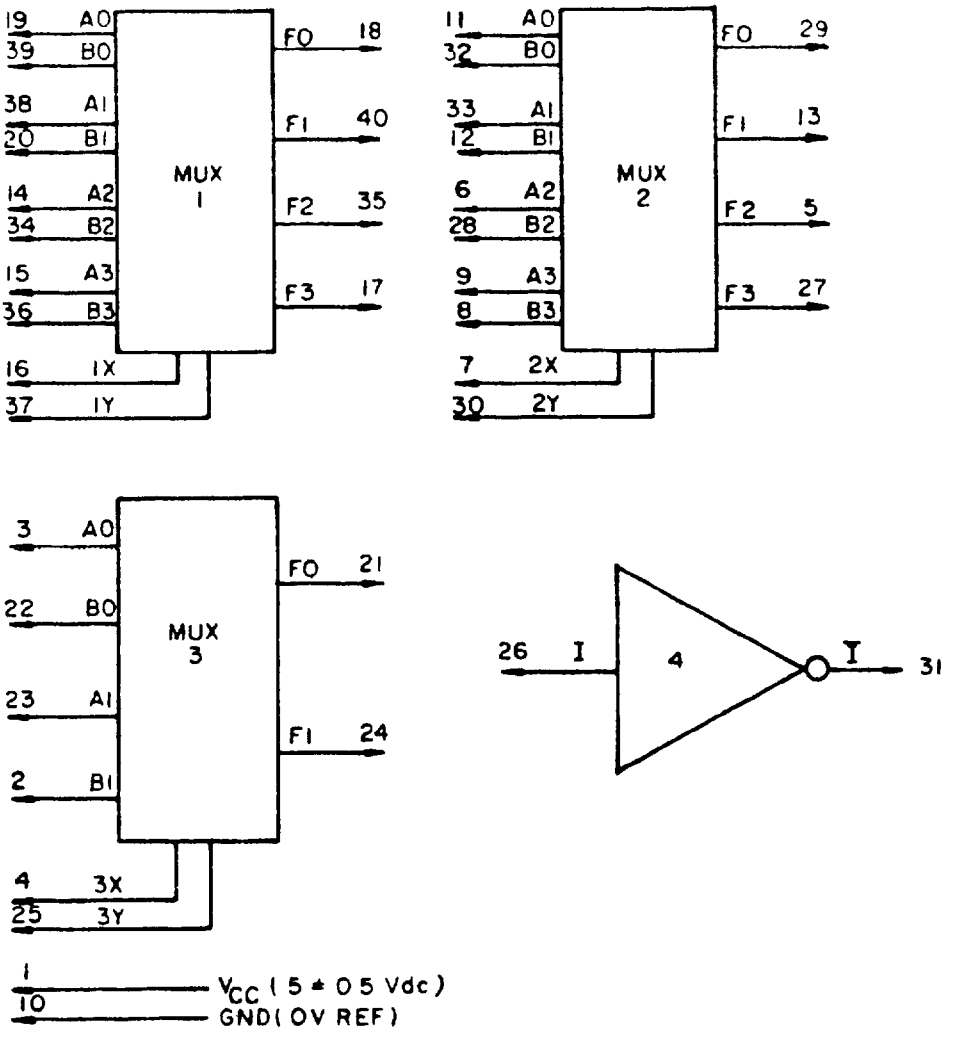


FIGURE 1 KDQ module function diagram, multiplexer

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1 A11	Input (low-loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1 B11	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
B14 B27	Input breakdown current	IV	S	0	2	0	2	0	2	0	2	mA
C	Isolation	V	S	10	-	1	-	10	-	1	-	M
D1 D11	Input capacitance	VI	Q	2	18	2	18	2	18	2	18	pF
D14-D25	Input capacitance	VI	Q	2	25	2	25	2	25	2	25	pF
D26,D27	Input capacitance	VI	Q	2	30	2	30	2	30	2	30	pF
E1 E27	Input clamp diode voltage	VII	Q	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1 5	V
F1-F72	Output voltage, high level	VIII	X	2 4	4.5	2 4	4.5	2.4	4.5	2.4	4 5	V
G1-G101	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	mV
H1-H13	Input current, high level	X	X	0	30	0	30	0	40	0	40	uA
H14-H27	Input current, high level	X	X	0	60	0	60	0	80	0	80	uA
J1-J13	Input current, low level	XI	X	0.4	1.73	0.4	1.73	0 4	1.73	0.4	1 73	mA
J14-J27	Input current, low level	XI	X	0 8	3.46	0.8	3.46	0 8	3.46	0.8	3 46	mA
K1-K11	Output short circuit current	XII	X	20	75	20	75	20	75	20	75	mA
L1	Power supply current, low level	XIII	S	1	323	1	323	1	323	1	323	mA
L2	Power supply current, high level	XIII	S	1	224	1	224	1	224	1	224	mA
M1-M11	Transition time (t_{TLH})	XIV	S	4	45	4	45	4	50	4	50	ns
N1-N11	Transition time (t_{THL})	XIV	S	2	20	2	20	2	25	2	25	ns
P1-P20	Propagation delay time (t_{PHL})	XV	S	7	31	7	31	7	34	7	34	ns

See footnotes at end of table.

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* TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P21-P30	Propagation delay time (t_{PHL})	XV	S	10	35	10	35	10	38	10	38	ns
P31-P40	Propagation delay time (t_{PHL})	XV	S	9	35	9	35	9	38	9	38	ns
P41-P50	Propagation delay time (t_{PHL})	XV	S	10	42	10	42	10	44	10	44	ns
P51-P60	Propagation delay time (t_{PHL})	XV	S	9	35	9	35	9	38	9	38	ns
P61-P70	Propagation delay time (t_{PHL})	XV	S	10	42	10	42	10	44	10	44	ns
P71	Propagation delay time (t_{PHL})	XV	S	2	19	2	19	2	22	2	22	ns
P1-R20	Propagation delay time (t_{PLH})	XV	S	7	26	7	26	7	28	7	28	ns
R21-R40	Propagation delay time (t_{PLH})	XV	S	10	35	10	35	10	38	10	38	ns
R41-R50	Propagation delay time (t_{PLH})	XV	S	10	42	10	42	10	44	10	44	ns
R51-R60	Propagation delay time (t_{PLH})	XV	S	10	35	10	35	10	38	10	38	ns
R61-R70	Propagation delay time (t_{PLH})	XV	S	10	42	10	42	10	44	10	44	ns
R71	Propagation delay time (t_{PLH})	XV	S	4	28	4	28	4	30	4	30	ns
S	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	mV
T1-T11	Output crosstalk, low level	XVII	Q	0	800	0	800	0	800	0	800	mV
T12-T15	Output crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	mV
T16-T21	Output crosstalk, low level	XVII	Q	0	450	0	450	0	450	0	450	mV
U1-U11	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply transient rise time	XIX	S	1	2	1	2	1	2	1	2	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through R and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limit at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/60 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10007-2658641. Part number M28787/60-1 supersedes part numbers 2658641 and M28787/60.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage	5 Vdc ±10 percent
Maximum power dissipation	
All outputs at low level	1.8 W (outputs not loaded)
All outputs at high level	1.3 W (outputs not loaded)
Input threshold voltage	
Low level	0.8 Vdc maximum
High level	2.0 Vdc maximum

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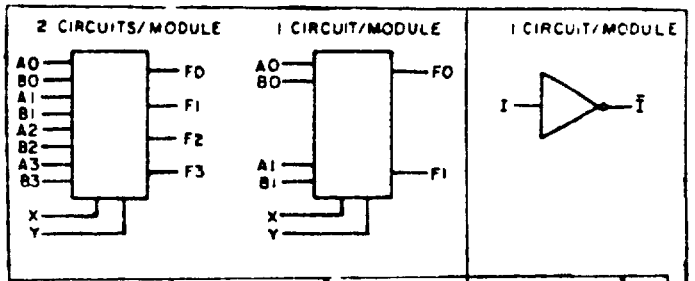
Input current	
Low level (V_{IL} = 0 V)	
A, 1, 3Y, 3Y	-1.73 mA maximum
B, 1X, 2X, 1Y, 2Y	-3.46 mA maximum
High level (V_{IH} = 2.4 V)	
A, 1, 3Y, 4Y	40 µA maximum
B, 1X, 1Y, 2X, 2Y	80 µA maximum
Output voltage	
Low level	0 Vdc minimum, 0.4 Vdc maximum
High level	2.4 Vdc minimum, 4.5 Vdc maximum
Output drive	
Low level (V_{OL} = 0 V)	16 mA maximum
High level (V_{OH} = 2.4 V)	-800 µA minimum
Propagation delay time (A-to-output)	
t_{PHL}	34 ns maximum
t_{PLH}	28 ns maximum
Propagation delay time (B-to-output, noninverting)	
t_{PHL}	34 ns maximum
t_{PLH}	28 ns maximum
Propagation delay time (B-to-output, inverting)	
t_{PHL}	38 ns maximum
t_{PLH}	38 ns maximum
Propagation delay time (X-address-to-output, noninverting)	
t_{PHL}	44 ns maximum
t_{PLH}	44 ns maximum
Propagation delay time (X-address-to-output, inverting)	
t_{PHL}	38 ns maximum
t_{PLH}	38 ns maximum
Propagation delay time (Y-address-to-output, noninverting)	
t_{PHL}	44 ns maximum
t_{PLH}	44 ns maximum
Propagation delay time (Y-address-to-output, inverting)	
t_{PHL}	38 ns maximum
t_{PLH}	38 ns maximum
Propagation delay time (inverter)	
t_{PHL}	22 ns maximum
t_{PLH}	30 ns maximum
Speed	DC to 20 MHz

* 6.5 Module function. This module contains four circuits, two circuits consist of four 2-way multiplexers with common select lines, one circuit consists of two 2-way multiplexers with common select lines, and one circuit consists of one independent inverter.

6.5.1 Input to output relationship. Each module has the input to output relationship of table II for positive logic.

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TABLE II Input to output relationship.



X-CONTROL	Y-CONTROL	F-OUTPUT	I	Ī
L	L	L	L	H
L	H	B		
H	L	B̄	H	L
H	H	A		

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Crosstalk relationships. The module has the following crosstalk relationships.

6.5.3.1 Module crosstalk. When connected in the worst case mode of operation (all but one output switching simultaneously from a high level to a low level) the KDQ module could exhibit substantial crosstalk. When the quiet output is a low level there may be as much as a 800 mV positive going spike coincident with the negative edge of the switching outputs.

6.5.3.2 Circuit crosstalk. Multiplexer circuits 2 and 3 can induce as much as 400 mV of noise onto any multiplexer circuit 1 output. Multiplexer circuits 1 and 3 can induce as much as 450 mV of noise onto any multiplexer circuit 2 output. Multiplexer circuits 1 and 2 can induce as much as 450 mV of noise onto any multiplexer 3 output.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

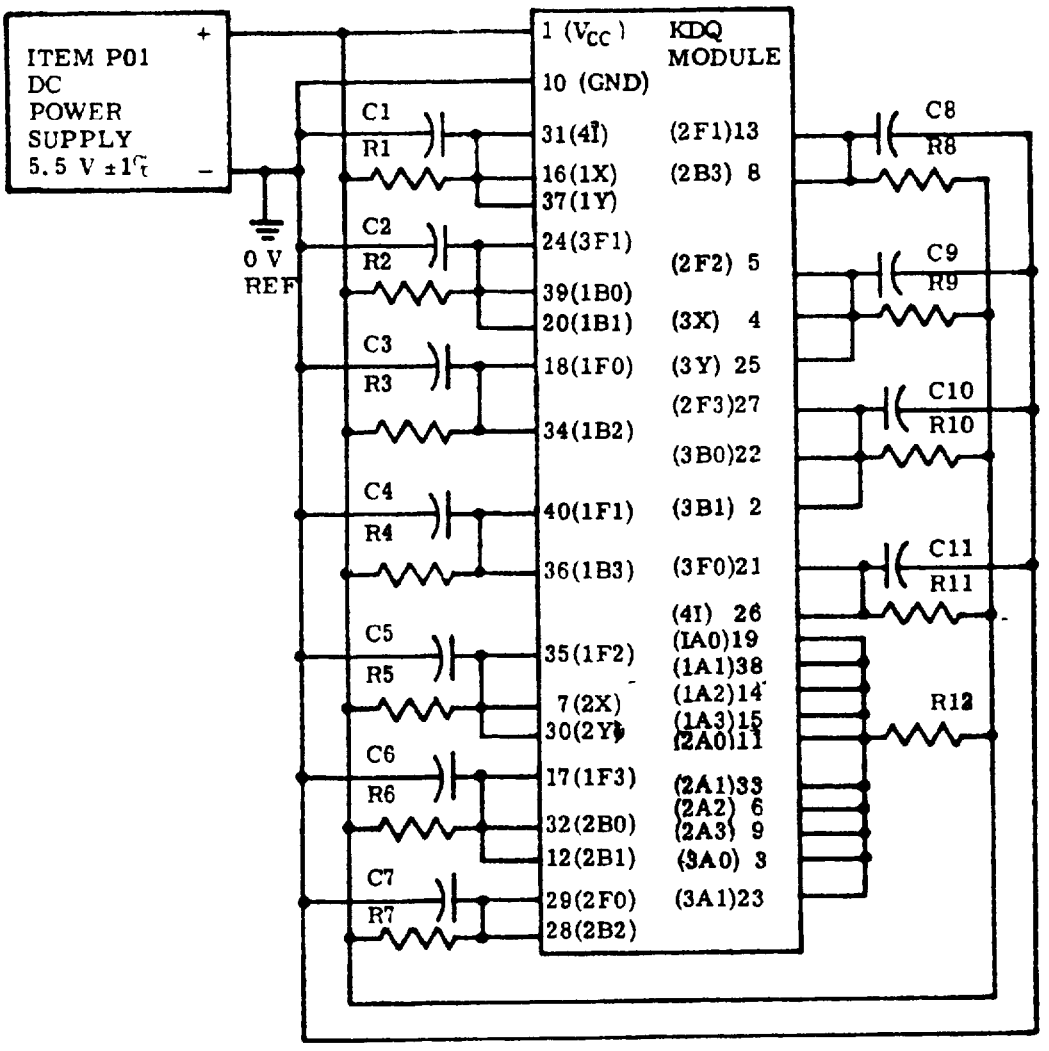
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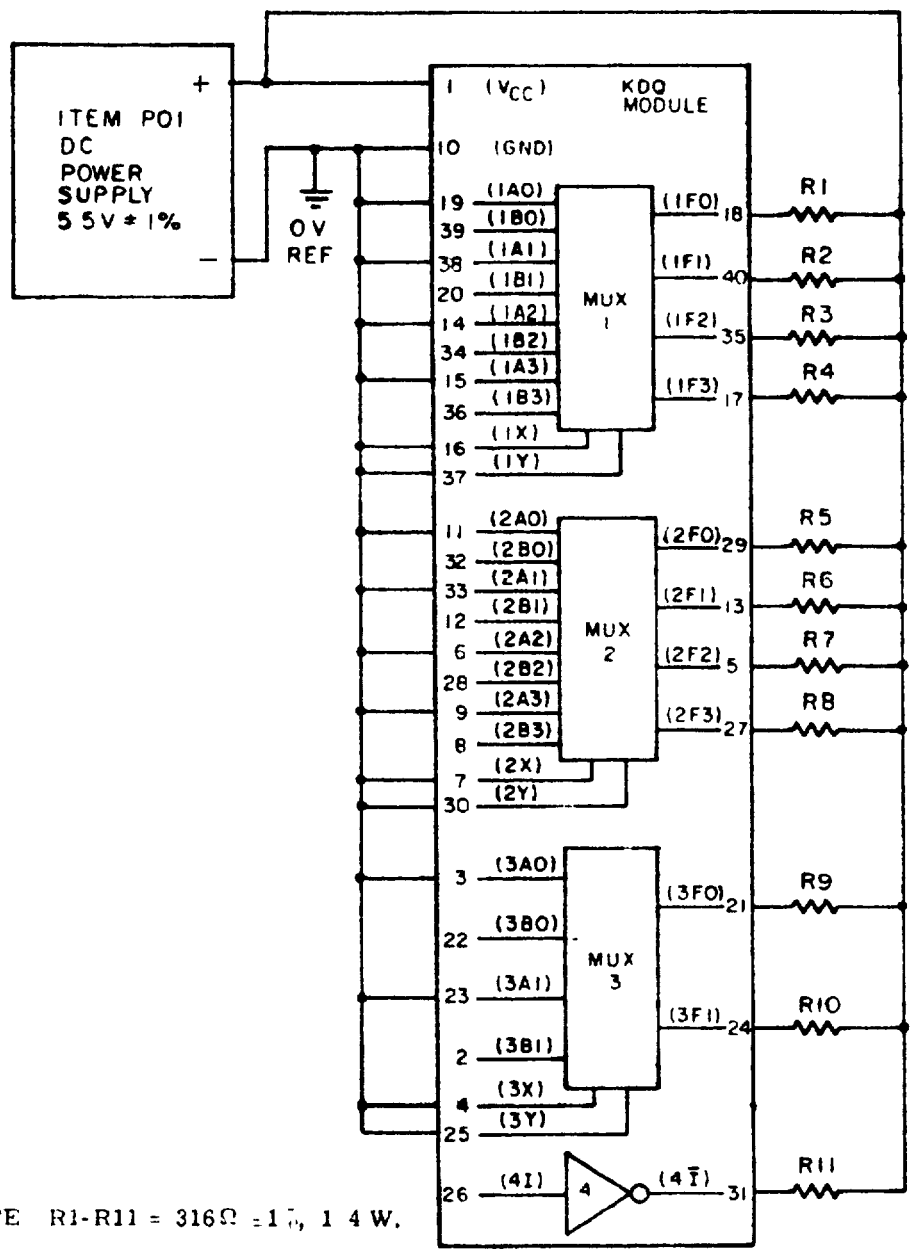
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NOTES
1 R1-R11 = 536Ω ±1%, 1/4 W
2 R12 = 1 kΩ ±5%, 1/4 W.
3 C1-C11 = 100 pF ±20%.

FIGURE 2 Operating temperature and life test circuit

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NOTE R1-R11 = 316Ω ±1%, 1/4 W.

FIGURE 3 Component temperature test circuit.

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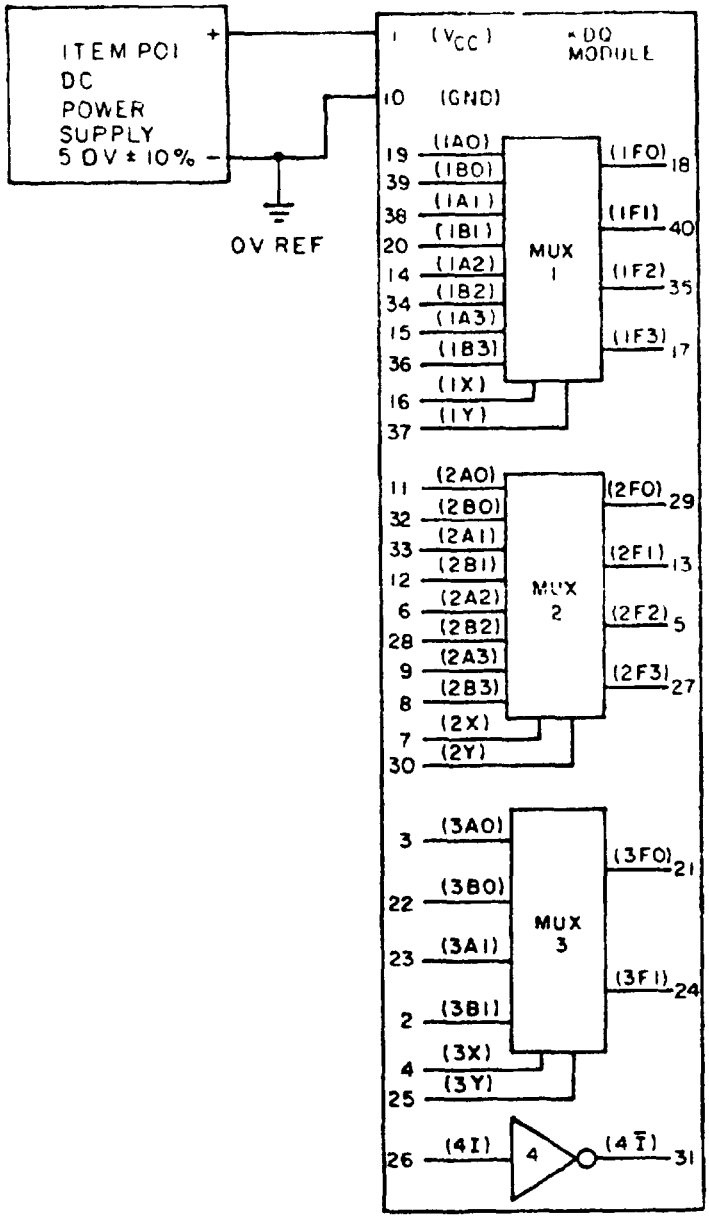


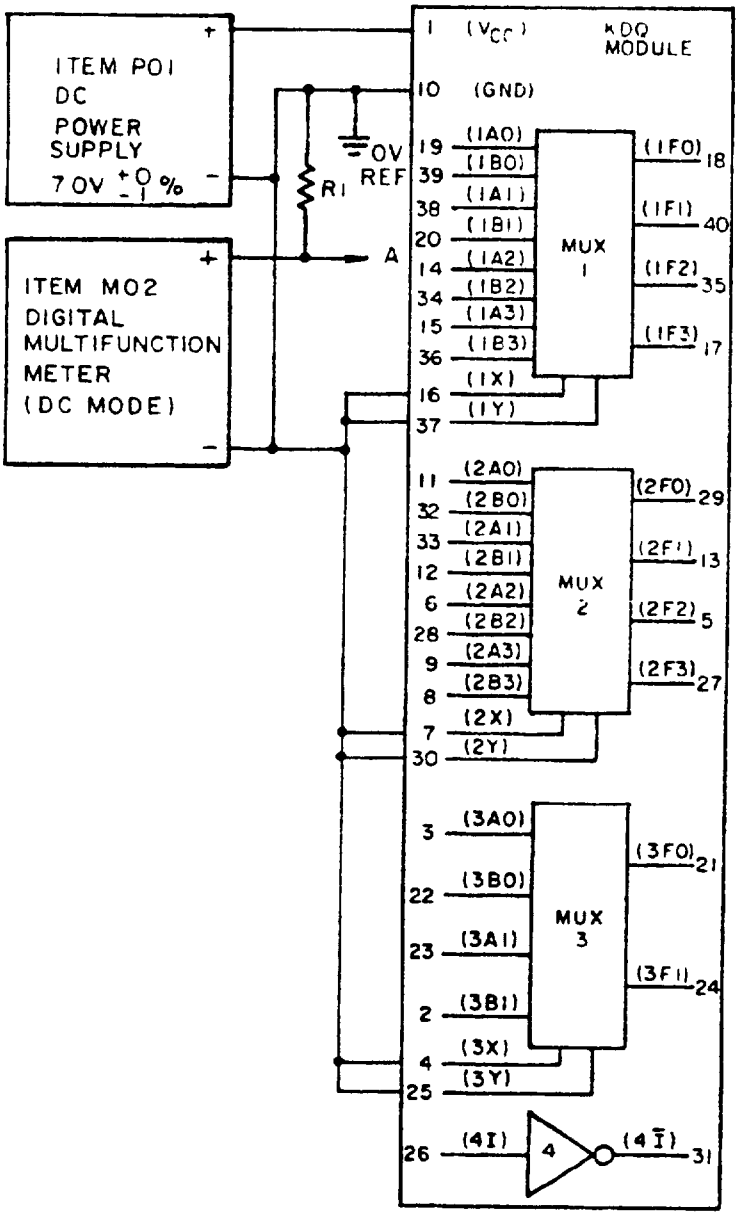
FIGURE 4 Vibration test circuit

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* TABLE III. V_{OUT} (low-loaded) test.

Test conditions	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1F0 V_{OUT} (low loaded)	18	10	800	mV	5
A2	1F1	40	↓	↓	↓	↓
A3	1F2	35				
A4	1F3	17				
A5	2F0	29				
A6	2F1	13				
A7	2F2	5				
A8	2F3	27				
A9	3F0	21				
A10	3F1	24				
A11	$4\overline{I}$	31				

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NOTE R1 = 316 Ω, 1/4 W

* FIGURE 5 V_{OU} - (low-loaded) test circuit for table I

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TABLE IV. Input breakdown current test.

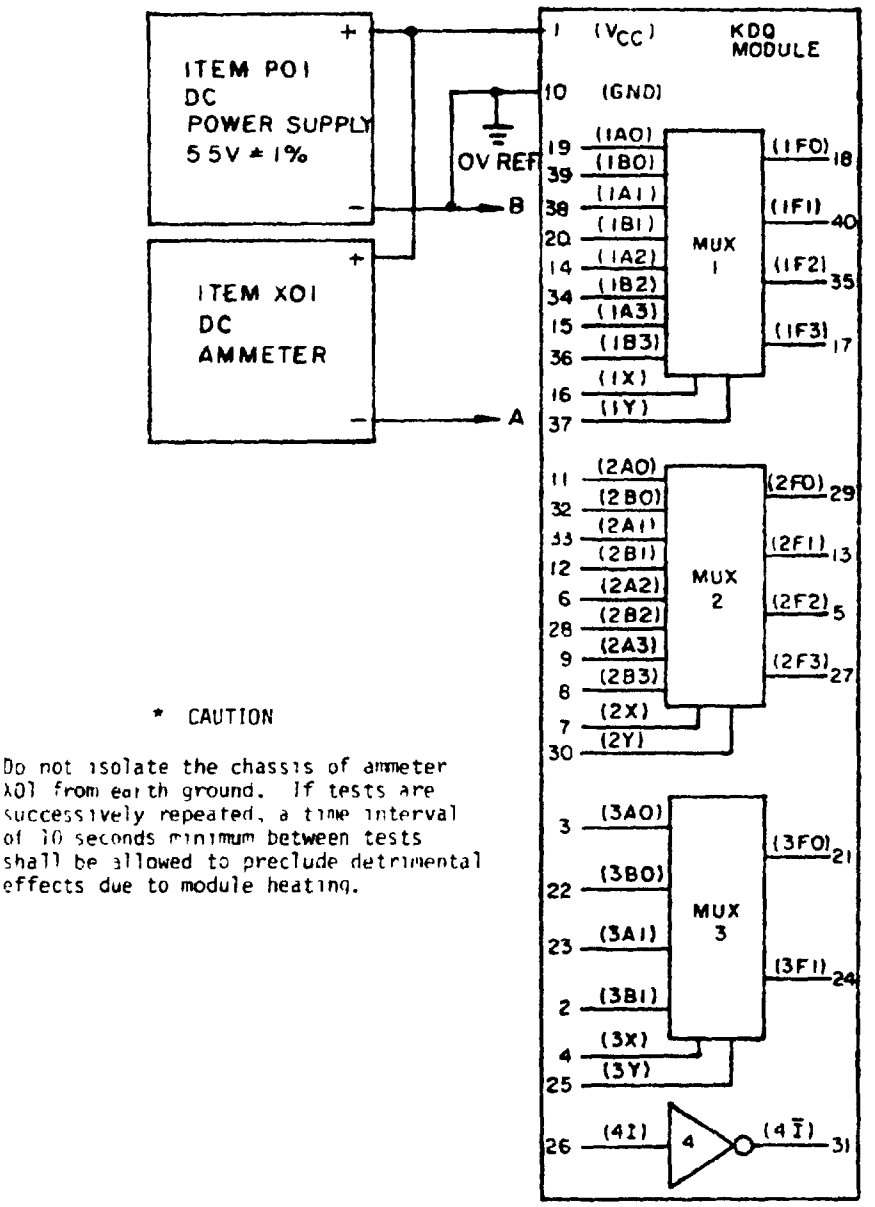
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A0 Input breakdown current	19	16,37	0	1	mA	6
B2	1A1	38					
B3	1A2	14					
B4	1A3	15					
B5	2A0	11	7,30				
B6	2A1	33					
B7	2A2	6					
B8	2A3	9					
B9	3A0	3	4,25				
B10	3A1	23	4,25				
B11	3X	4	Open				
B12	3Y	25	Open				
B13	4I	26	Open				
B14	1B0	39	37		2		
B15	1B1	20					
B16	1B2	34					
B17	1B3	36					
B18	1X	16	Open				
B19	1Y	37					
B20	2X	7					
B21	2Y	30					
B22	2B0	32	30				
B23	2B1	12					
B24	2B2	28					

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TABLE IV Input breakdown ent test - Continued

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B25	2B3 Input breakdown current	8	30	0	2	mA	6
B26	3B0 Input breakdown current	22	25	0	2	mA	6
B27	3B1 Input breakdown current	2	25	0	2	mA	6

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* CAUTION

Do not isolate the chassis of ammeter X01 from earth ground. If tests are successively repeated, a time interval of 10 seconds minimum between tests shall be allowed to preclude detrimental effects due to module heating.

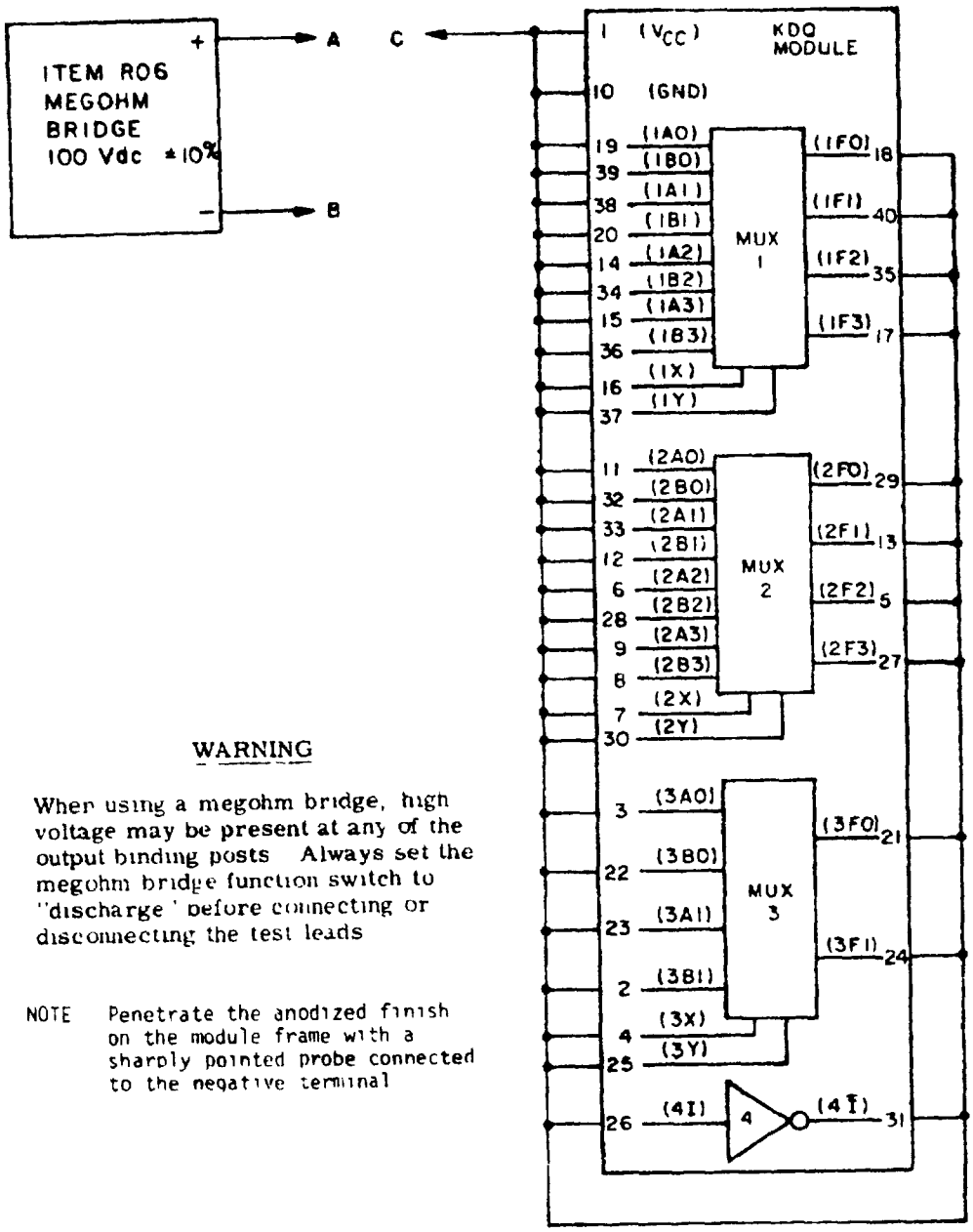
FIGURE 6 Input breakdown current test circuit for table IV

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* TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	C	Frame	10	∞	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 7. Isolation test circuit for table V.

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TABLE VI Input capacitance test

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figur
01	1A2 Input capacitance	14	2	18	pF	8
02	1A3	15				
03	1A0	19				
04	1A1	38				
05	2A2	6				
06	2A3	9				
07	2A0	11				
08	2A1	33				
09	3A0	3				
010	3X	4				
011	3A1	23				
012	3Y	25				
013	4I	26				
014	1B1	20		25		
015	1B2	34				
016	1B3	36				
017	1L0	39				
018	1Y	37				
019	2B3	8				
020	2B1	12				
021	2B2	28				
022	2Y	30				
023	2B0	32				
024	3B1	2				

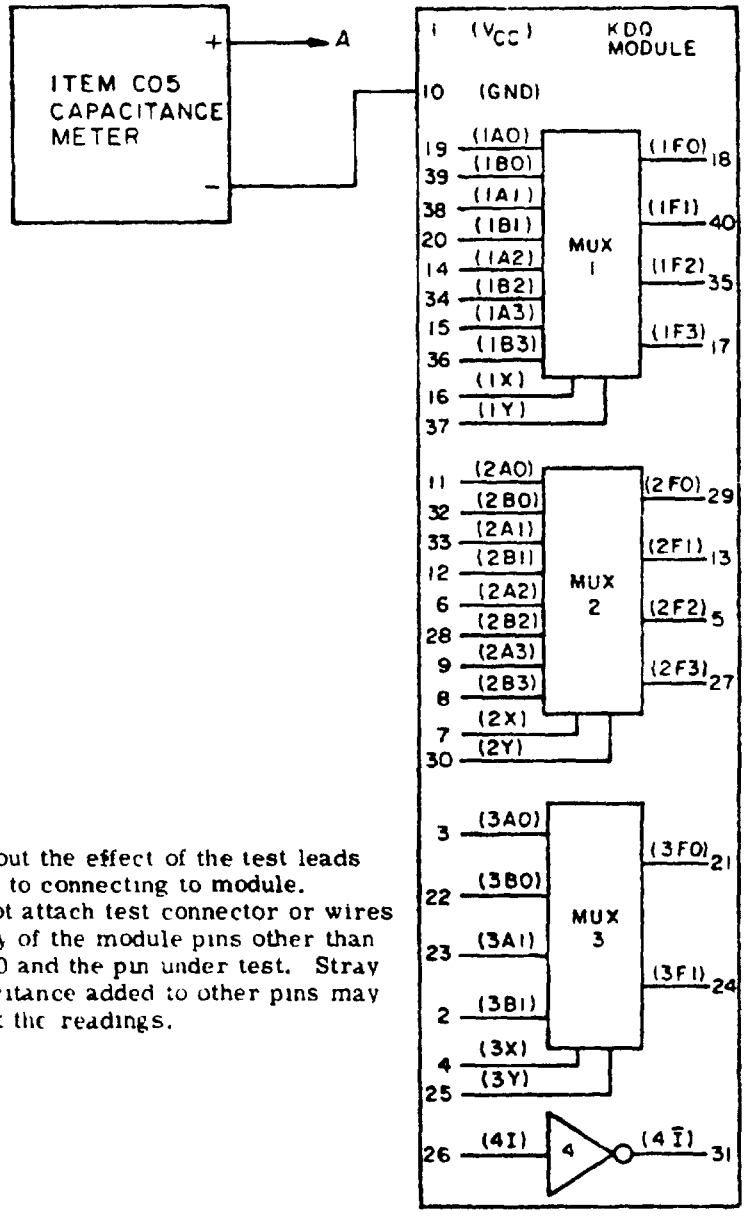
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TABLE VI. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D.C	3B0 Input capacitance	22	2	25	pF	8
D.C	1X Input capacitance	16	2	30	pF	8
	2X Input capacitance	7	2	30	pF	8

MIL-M-28787/4(SH)

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NOTES
1 Null out the effect of the test leads prior to connecting to module.
2 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

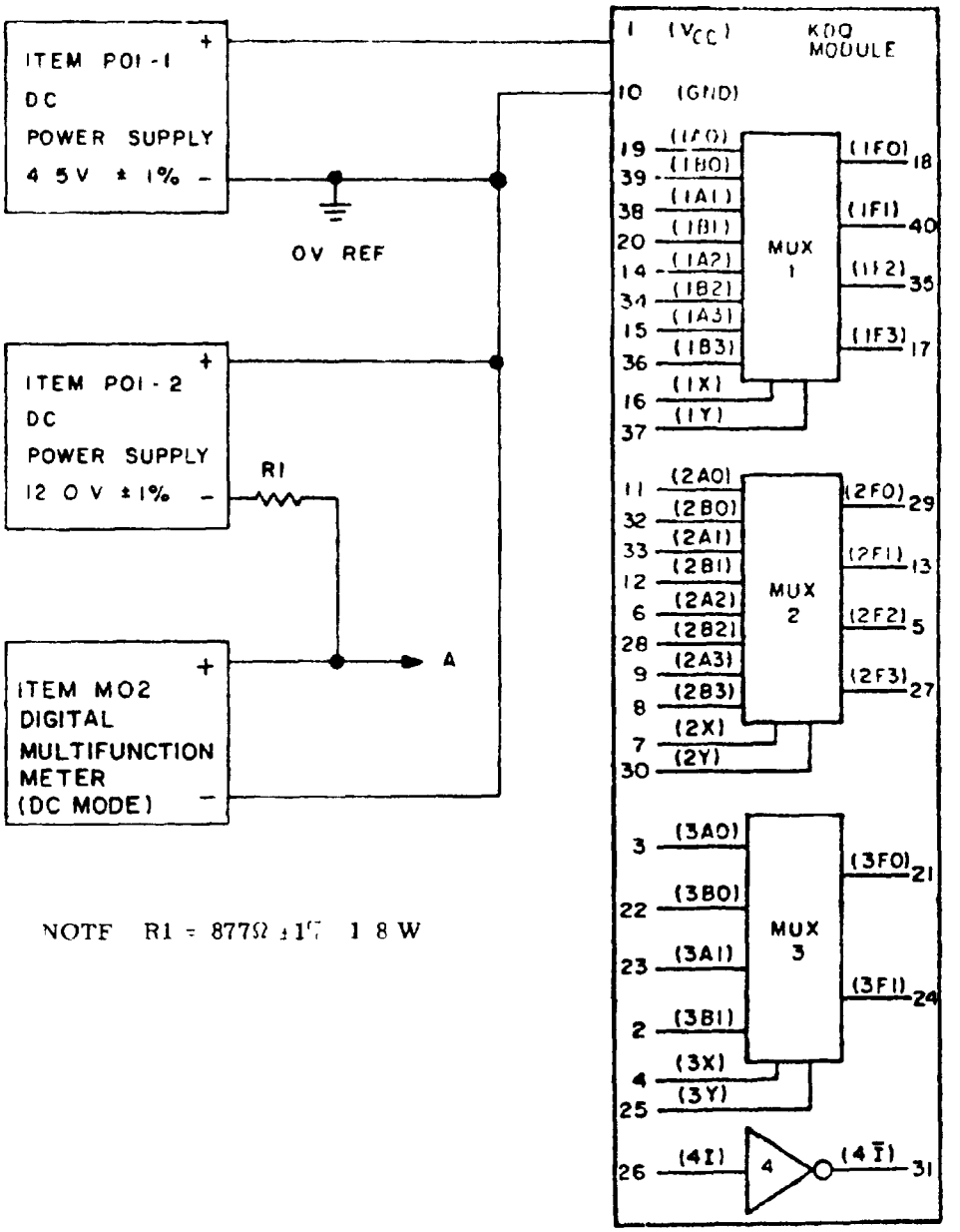
FIGURE 8 Input capacitance test circuit for table VI

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TABLE VII Input clamp diode voltage test

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
	1A0 Input clamp diode voltage	19	-0.4	-1 5	V	9
	1A1	38				
	1A2	14				
	1A3	15				
	2A0	11				
	2A1	33				
	2A2	6				
	2A3	9				
	3A0	3				
	3A1	23				
E11	3X	4				
E12	3Y	25				
E13	4I	26				
E14	1B0	39				
	1B1	20				
	1B2	34				
	1B3	36				
	1X	16				
	1Y	37				
	2X	7				
	2Y	30				
	2B0	32				
	2B1	12				
	2B2	28				
	2B3					
	3B0					

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NOTE R1 = 877Ω ±1% 1 8 W

* FIGURE 9 Input clamp diode voltage test circuit for table VII

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* TABLE VIII Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1F3 Output voltage, high level	14,15,19,20, 34,36,37,38, 39	16	17	2.4	4.5	V	10
F2	1F0			18				
F3	1F2			35				
F4	1F1			40				
F5	1F3	20,34,36,37, 39	14,15,16, 19,38	17				
F6	1F0			18				
F7	1F2			35				
F8	1F1			40				
F9	1F3	14,15,16,19, 37,38	20,34,36, 39	17				
F10	1F0			18				
F11	1F2			35				
F12	1F1			40				
F13	1F3	14,15,16,19, 20,34,36,37, 38,39	Open	17				
F14	1F0			18				
F15	1F2			35				
F16	1F1			40				

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* TABLE VIII. Output voltage (high level) test - Continued

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F17	1F3 Output voltage, high level	14,15,16,19, 38	20,34,36, 37,39	17	2.4	4.5	V	10
F18	1F0			18				
F19	1F2			35				
F20	1F1			40				
F21	1F3	16	14,15,19, 20,34,36 37,38,39	17				
F22	1F0			18				
F23	1F2			35				
F24	1F1			40				
F25	2F2	6,8,9,11,12, 28,30,32,33	7	5				
F26	2F1			13				
F27	2F3			27				
F28	2F0			29				
F29	2F2	8,12,28,30, 32	6,7,9,11, 33	5				
F30	2F1			13				
F31	2F3			27				
F32	2F0			29				

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* TABLE VIII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F33	2F2 Output voltage, high level	6,7,9,11,30, 33	8,12,28,32,	5	2.4	4.5	V	10
F34	2F1			13				
F35	2F3			27				
F36	2F0			29				
F37	2F2	6,7,8,9,11, 12,28,30,32, 33	Open	5				
F38	2F1			13				
F39	2F3			27				
F40	2F0			29				
F41	2F2	6,7,9,11,33	8,12,28,30, 32	5				
F42	2F1			13				
F43	2F3			27				
F44	2F0			29				
F45	2F2	7	6,8,9,11, 12,28,30, 32,33	5				
F46	2F1			13				
F47	2F3			27				
F48	2F0			29				

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* TABLE VIII Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F49	3F0 Output voltage, high level	2,3,22,23, 25	4	21	2 4	4 5	V	10
F50	3F1	2,3,22,23, 25	4	24				
F51	3F0	2,22,25	3,4,23	21				
F52	3F1	2,22,25	3,4,23	24				
F53	3F0	3,4,23,25	2,22	21				
F54	3F1	3,4,23,25	2,22	24				
F55	3F0	2,3,4,22,23, 25	Open	21				
F56	3F1	2,3,4,22,23, 25	Open	24				
F57	3F0	3,4,23	2,22,25	21				
F58	3F1	3,4,23	2,22,25	24				
F59	3F0	4	2,3,22,23, 25	21				
F60	3F1	4	2,3,22,23, 25	24				
F61	4$\overline{1}$	Open	26	31				
F62	1F0		4,7,25,30, 37,39	18				
F63	1F1		4,7,20,25, 30,37	40				
F64	1F2		4,7,25,30, 34,37	35				
F65	1F3		4,7,25,30, 36,37	17				
F66	3F0		7,16,22,25, 30,37	21				

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* TABLE VIII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F67	3F1 Output voltage, high level	Open	2,7,16,25, 30,37	24	2.4	4.5	V	10
F68	2F0		4,16,25, 30,32,37	29				
F69	2F1		4,12,16, 25,30,37	13				
F70	2F2		4,16,25, 28,30,37	5				
F71	2F3		4,8,16,25, 30,37	27				
F72	$4\bar{1}$		4,7,16,25, 26,30,37	31				

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* TABLE IX. Output voltage (low-level) test

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1F3 Output voltage, low level	14,15,19, 37,38	16,20,34, 36,39	17	0	400	mV	10
G2	1F0	↓	↓	18	↓	↓	↓	↓
G3	1F2			35				
G4	1F1			40				
G5	1F3	37	14,15,16, 19,20,26, 34,38,39	17				
G6	1F0		↓	18				
G7	1F2			35				
G8	1F1			40				
G9	1F3	16,20,34, 36,37,39	14,15,19, 38	17				
G10	1F0	↓	↓	18				
G11	1F2			35				
G12	1F1			40				
G13	1F3	16,37	14,15,19, 20,34,36, 38,39	17				
G14	1F0	↓	↓	18				
G15	1F2			35				
G16	1F1	↓	↓	40	↓	↓	↓	↓

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* TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G17	1F3 Output voltage, low level	14,15,16, 19,20,34, 36,38,39	37	17	0	400	mV	10
G18	1F0			18				
G19	1F2			35				
G20	1F1			40				
G21	1F3	16,20,34, 36,39	14,15,19, 37,38	17				
G22	1F0			18				
G23	1F2			35				
G24	1F1			40				
G25	1F3	14,15,19, 38	16,20.34, 37,39,36	17				
G26	1F0			18				
G27	1F2			35				
G28	1F1			40				
G29	1F3	14,15,19, 20,34,36, 38,39	16,37	17				
G30	1F0			18				
G31	1F2			35				
G32	1F1			40				

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* TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G33	1F3 Output voltage, low level	20,34,36, 39	14,15,16, 19,37,38	17	0	400	mV	10
G34	1F0			18				
G35	1F2			35				
G36	1F1			40				
G37	1F3	Open	14,15,16, 19,20,34, 36,37,38, 39	17				
G38	1F0			18				
G39	1F2			35				
G40	1F1			40				
G41	2F2	6,9,11, 30,33	7,8,12, 28,32	5				
G42	2F1			13				
G43	2F3			27				
G44	2F0			29				
G45	2F2	30	6,7,8,9, 11,12,28, 32,33	5				
G46	2F1			13				
G47	2F3			27				
G48	2F0			29				

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* TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G49	2F2 Output voltage, low level	7,8,12, 28,30,32	6,9,11,33,	5	0	400	mV	10
G50	2F1			13				
G51	2F3			27				
G52	2F0			29				
G53	2F2	7,30	6,8,9,11, 12,28,32, 33	5				
G54	2F1			13				
G55	2F3			27				
G56	2F0			29				
G57	2F2	6,7,8,9, 11,12,28, 32,33	30	5				
G58	2F1			13				
G59	2F3			27				
G60	2F0			29				
G61	2F2	7,8,12, 28,32	6,9,11, 30,33	5				
G62	2F1			13				
G63	2F3			27				
G64	2F0			29				

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* TABLE IX Output voltage (low-level) test - Continued.

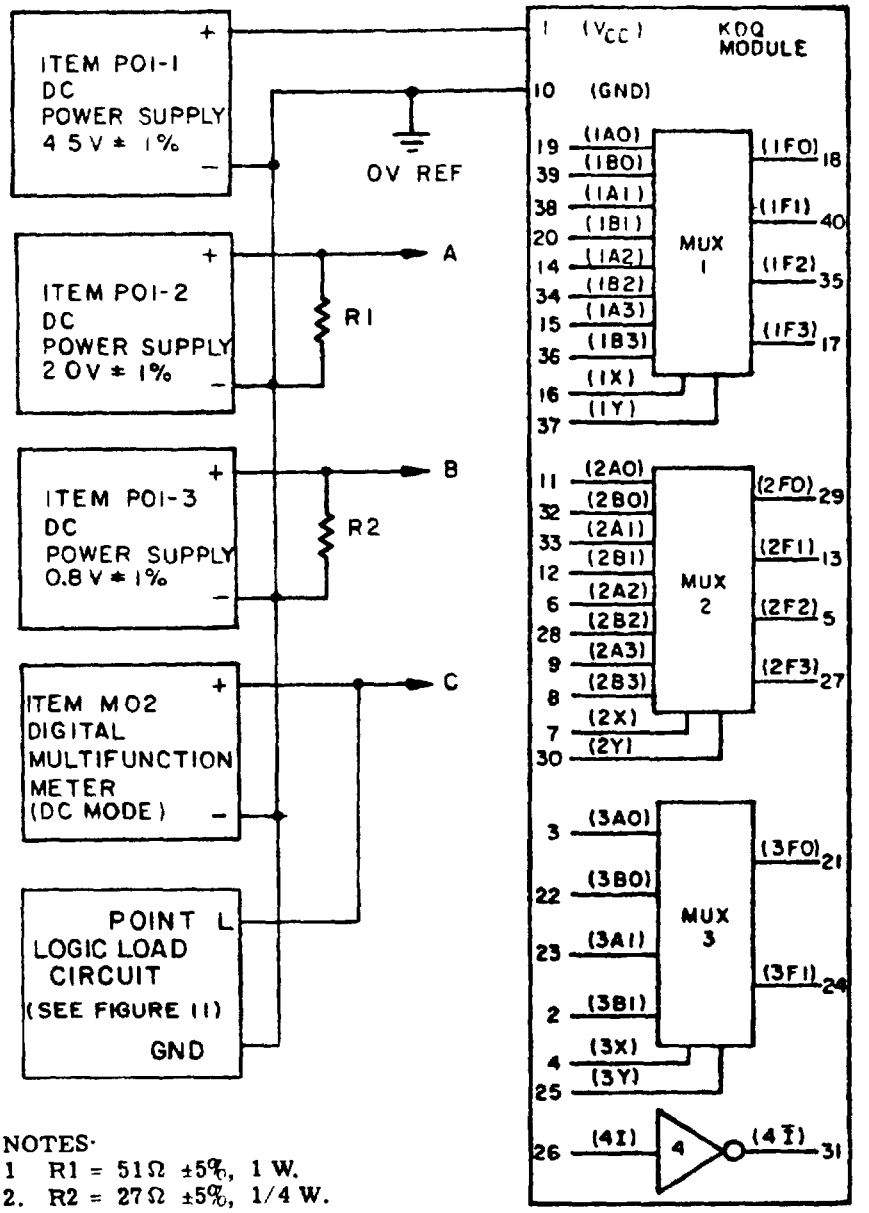
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G65	2F2 Output voltage, low level	6,9,11,33,	7,8,12, 28,30,32	5	0	400	mV	10
G66	2F1			13				
G67	2F3			27				
G68	2F0			29				
G69	2F2	6,8,9,11, 12,28,32, 33	7,30	5				
G70	2F1			13				
G71	2F3			27				
G72	2F0			29				
G73	2F2	8,12,28,32	6,7,9,11, 30,33	5				
G74	2F1			13				
G75	2F3			27				
G76	2F0			29				
G77	2F2	Open	6,7,8,9, 11,12,28, 30,32,33	5				
G78	2F1			13				
G79	2F3			27				
G80	2F0			29				

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* TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G81	3F0 Output voltage, low level	3,23,25	2,4,22	21	0	400	mV	10
G82	3F1	3,23,25	2,4,22	24				
G83	3F0	25	2,3,4,22, 23	21				
G84	3F1	25	2.3.4.22, 23	24				
G85	3F0	2,4,22,25	3,23	21				
G86	3F1	2,4,22,25	3,23	24				
G87	3F0	4,25	2,3,22,23	21				
G88	3F1	4,25	2,3,22,23	24				
G89	3F0	2,3,4,22,23	25	21				
G90	3F1	2,3,4,22,23	25	24				
G91	3F0	2,4,22	3,23,25	21				
G92	3F1	2,4,22	3,23,25	24				
G93	3F0	3,23	2,4,22,25	21				
G94	3F1	3,23	2,4,22,25	24				
G95	3F0	2,3,22,23	4,25	21				
G96	3F1	2,3,22,23	4,25	24				
G97	3F0	2,22	3,4,23,25	21				
G98	3F1	2,22	3,4,23,25	24				
G99	3F0	Open	2,3,4,22, 23,25	21				
G100	3F1	Open	2,3,4,22, 23,25	24				
G101	4$\bar{I}$	26	Open	31				

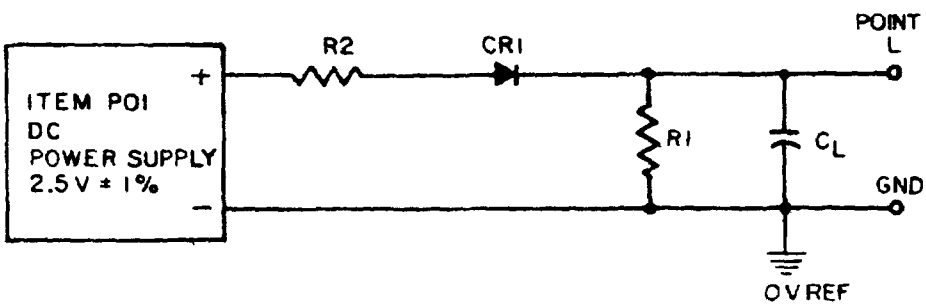
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 27Ω ±5%, 1/4 W.

★ FIGURE 10. Output voltage (high-level and low-level) test circuit for tables VIII and IX.

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NOTES:
1. For test conditions M, N, P and R, C_L = 100 ±5 pF (including probe and parasitic capacitance).
2. CR1 - 1N4150, or equivalent
3. R1 = 3 01 kΩ ±1%, 1/8 W
4. R2 = select value for current flow of 17 3 mA ±1% from point L to ground.
5. C_L is not required for dc electrical test conditions F and G (table II).
6. The maximum lead length from point L to the appropriate module pin shall be less than 1 inch (25 4 mm).
7. During output switching, the 2. 5 V power supply shall have a maximum overshoot of 500 mV.
8. For test conditions T and U, C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 11 Logic load circuit

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TABLE X Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1A0 Input current, high level	19	Open	16,37	0	30	µA	12
H2	1A1	38						
H3	1A2	14						
H4	1A3	15						
H5	2A0	11		7,30				
H6	2A1	33						
H7	2A2	6						
H8	2A3	9						
H9	3A0	3		4,25				
H10	3A1	23		4,25				
H11	3X	4		Open				
H12	3Y	25		Open				
H13	$4\overline{1}$	26		Open				
H14	1B0	39		37		60		
H15	1B1	20						
H16	1B2	34						
H17	1B3	36						
H18	2B0	32		30				
H19	2B1	12						
H20	2B2	28						
H21	2B3	8						
H22	3B0	22		25				
H23	3B1	2		25				
H24	1X	16		Open				

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TABLE A Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H25	1Y Input current, high level	37	Open	Open	0	60	µA	12
H26	2X Input current, high level	7	Open	Open	0	60	µA	12
H27	2Y Input current, high level	30	Open	Open	0	60	µA	12

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TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	1A0 Input current, low level	Open	19	Open	0.4	1.73	mA	12
J2	1A1		38					
J3	1A2		14					
J4	1A3		15					
J5	2A0		11					
J6	2A1		33					
J7	2A2		6					
J8	2A3		9					
J9	3A0		3					
J10	3A1		23					
J11	3X		4					
J12	3Y		25					
J13	4I		26					
J14	1B0		39	16	0.8	3.46		
J15	1B1		20					
J16	1B2		34					
J17	1B3		36					
J18	2B0		32	7				
J19	2B1		12					
J20	2B2		28					
J21	2B3		8					
J22	3B0		22	4				
J23	3B1		2	4				
J24	1X		16	Open				

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TABLE XI Input current (low-level) test - Continued.

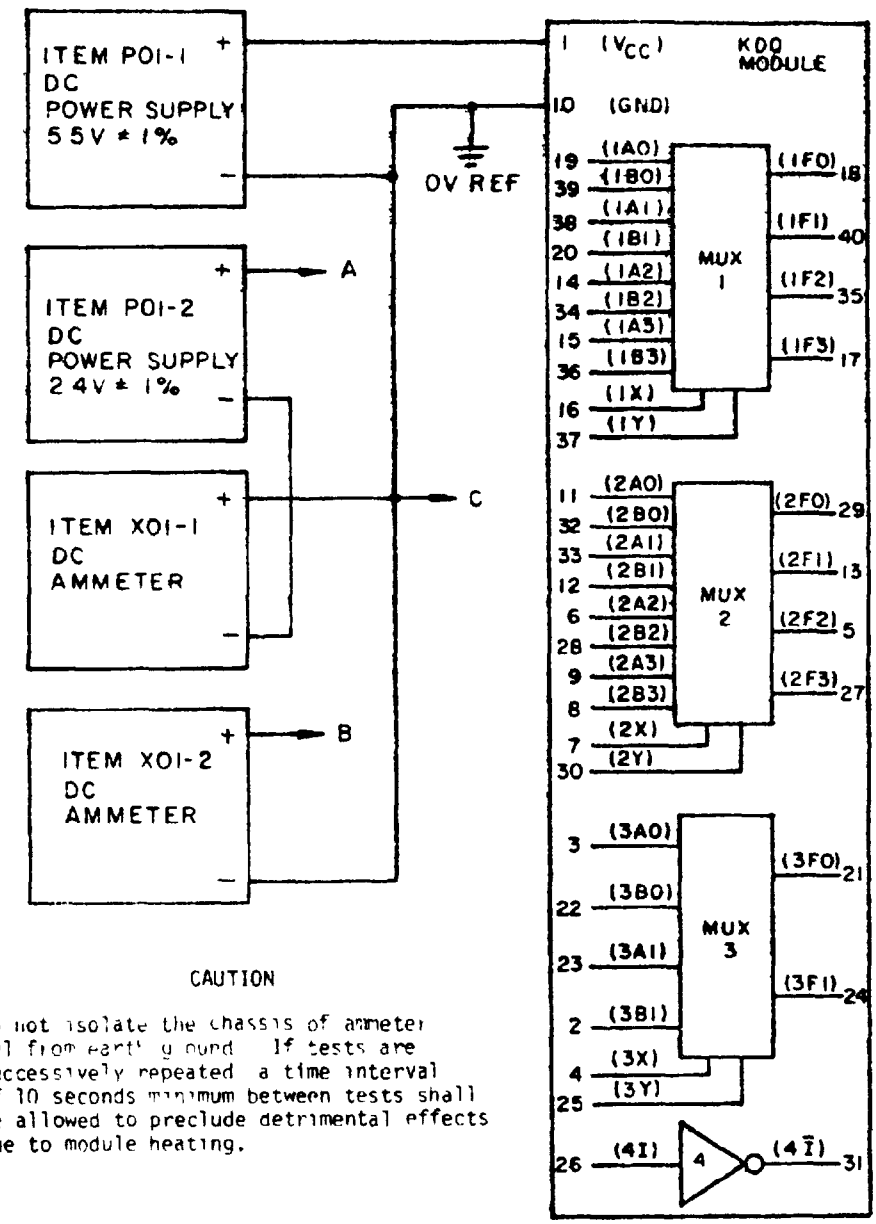
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J25	1Y Input current, low level	Open	37	Open	0.8	3.46	mA	12
J26	2X Input current, low level	Open	7	Open	0.8	3.46	mA	12
J27	2Y Input current, low level	Open	30	Open	0.8	3.46	mA	12

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TABLE XII Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1F0 Output short circuit current	Open	18	Open	20	75	mA	12
K2	1F1		40					
K3	1F2		35					
K4	1F3		17					
K5	2F0		29					
K6	2F1		13					
K7	2F2		5					
K8	2F3		27					
K9	3F0		21					
K10	3F1		24					
K11	$4\bar{1}$		31	26				

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NOTES

1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01, is required. Use item X01-2 for tables XI and XII, use item X01-1 for table X.
2. With point "A" disconnected, item X01-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

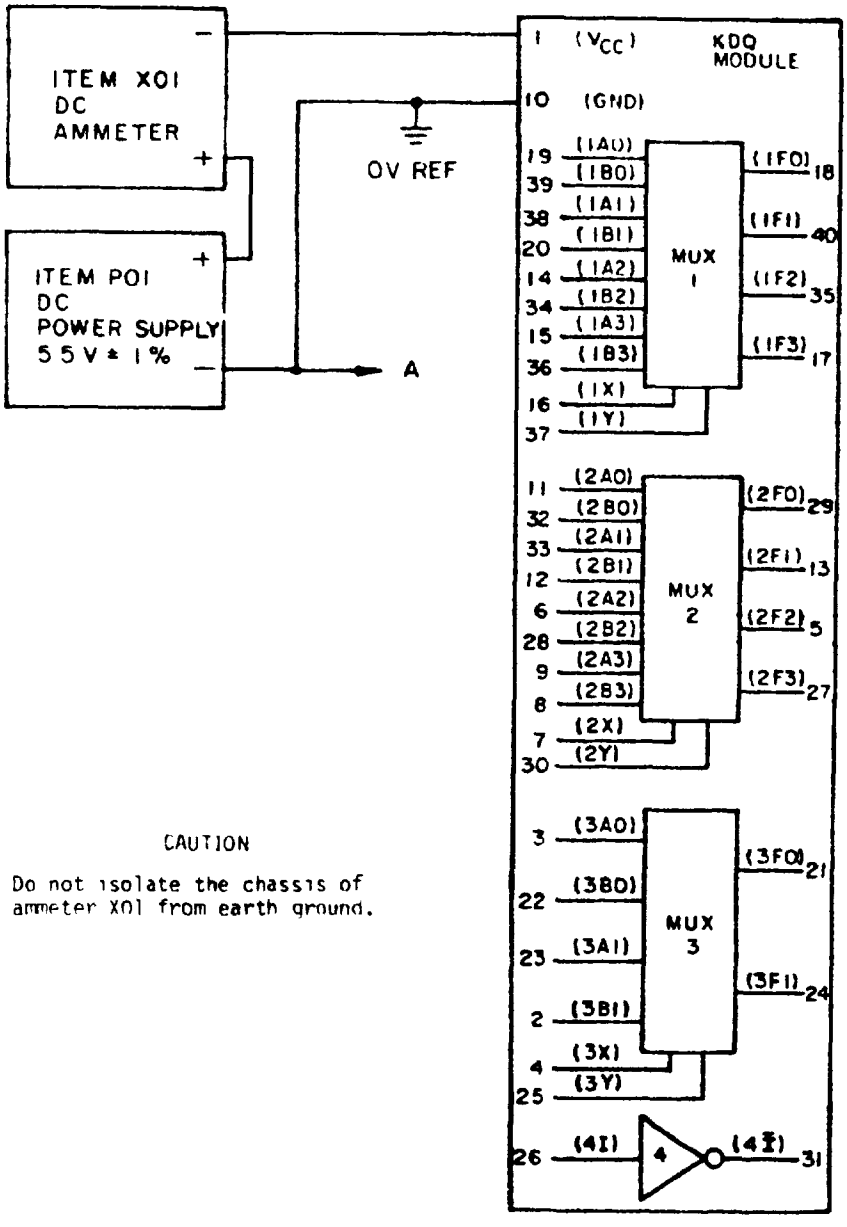
* FIGURE 12. Input current (high-level and low-level) and output short circuit current test circuit for tables X, XI, and XII.

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TABLE XIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Power supply current, low level	19,38,16,37,14,15,11, 33,7,30,6,9,3,23,4,25	1	323	mA	13
L2	Power supply current, high level	39,20,34,36,32,12,28, 8,22,2,26	1	224	mA	13

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* FIGURE 13 Power supply current (low-level and high-level) test circuit for table XIII

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* TABLE XIV Transition time (t_{TLH} and t_{THL}) test

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1F0 Transition time (t_{TLH})	19	16,37	39,20	18	4	45	ns	14,1
M2	1F1	38		39,20	40				
M3	1F2	14		34,36	35				
M4	1F3	15		34,36	17				
M5	2F0	11	7,30	32,12	29				
M6	2F1	33		32,12	13				
M7	2F2	6		28,8	5				
M8	2F3	9		28,8	27				
M9	3F0	3	4,25	22,2	21				
M10	3F1	23	4,25	22,2	24				
M11	4F	26	Open	Open	31				

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* TABLE XIV Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1F0 Transition time (t_{THL})	19	16,37	39,20	18	2	20	ns	14,15
N2	1F1	38		39,20	40				
N3	1F2	14		34,36	35				
N4	1F3	15		34,36	17				
N5	2F0	11	7,30	32,12	29				
N6	2F1	33		32,12	13				
N7	2F2	6		28,8	5				
N8	2F3	9		28,8	27				
N9	3F0	3	4,25	22,2	21				
N10	3F1	23	4,25	22,2	24				
N11	4$\bar{1}$	26	Open	Open	31				

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test

Test condition	Parameter	Wave form	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1F0 Propagation delay time (t_{PHL})	1	19	16,37	39,20	18	7	31	ns	4,15
P2	1F1		38		39,20	40				
P3	1F2		14		34,36	35				
P4	1F3		15		34,36	17				
P5	2F0		11	7,30	32,12	29				
P6	2F1		33		32,12	13				
P7	2F2		6		28,8	5				
P8	2F3		9		28,8	27				
P9	3F0		3	4,25	22,2	21				
P10	3F1		23	4,25	22,2	24				
P11	1F0		39	37	19,38,16	18				
P12	1F1		20		19,38,16	40				
P13	1F2		34		14,15,16	35				
P14	1F3		36		14,15,16	17				
P15	2F0		32	30	7,11,33	29				
P16	2F1		12		7,11,33	13				
P17	2F2		28		7,6,9	5				
P18	2F3		8		7,6,9	27				
P19	3F0		22	25	3,23,4	21				
P20	3F1		2	25	3,23,4	24				
P21	1F0	2	39	16	19,38,37	18	10	35		
P22	1F1		20		19,38,37	40				
P23	1F2		34		14,15,37	35				
P24	1F3		36		14,15,37	17				

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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Wave-form	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P25	2F0 Propagation delay time (t_{PHL})	2	32	7	11,33,30	29	10	35	ns	14,15
P26	2F1		12		11,33,30	13				
P27	2F2		28		6,9,30	5				
P28	2F3		8		6,9,30	27				
P29	3F0		22	4	3,23,25	21				
P30	3F1		2	4	3,23,25	24				
P31	1F0		16	37,39,20	19,38	18	9			
P32	1F1			37,39,20	19,38	40				
P33	1F2			37,34,36	14,15	35				
P34	1F3			37,34,36	14,15	17				
P35	2F0		7	30,32,12	11,33	29				
P36	2F1			30,32,12	11,33	13				
P37	2F2			30,28,8	6,9	5				
P38	2F3			30,28,8,	6,9	27				
P39	3F0		4	25,22,2	3,23	21				
P40	3F1		4	25,22,2	3,23	24				
P41	1F0	1	16	Open	39,19,20, 38,37	18	10	42		
P42	1F1				39,19,20, 38,37	40				
P43	1F2				14,34,15, 36,37	35				
P44	1F3				14,34,15, 36,37	17				
P45	2F0		7		11,32,33, 12,30	29				

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Test condition	Parameter	Wave form	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P46	2F1 Propagation delay time (t_{PHL})	1	7	Open	11,32,33, 12,30	13	10	42	ns	14,15
P47	2F2				6,28,9, 8,30	5				
P48	2F3				6,28,9, 8,30	27				
P49	3F0		4		3,22,23, 2,25	21				
P50	3F1		4		3,22,23, 2,25	24				
P51	1F0	2	37	16	19,39,38, 20	18	9	35		
P52	1F1				19,39,38, 20	40				
P53	1F2				14,34,15, 36	35				
P54	1F3				14,34,15, 36	17				
P55	2F0		30	7	11,32,33, 12	29				
P56	2F1				11,32,33, 12	13				
P57	2F2				6,28,9,8,	5				
P58	2F3				6,28,9,8,	27				
P59	3F0		25	4	3,22,23, 2	21				
P60	3F1		25	4	3,22,23, 2	24				
P61	1F0	1	37	39,20	19,38,16	18	10	42		

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Test condition	Parameter	Wave-form	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P62	1F1 Propagation delay time (t_{PHL})	1	37	39,20	19,38,16	40	10	42	ns	14,15
P63	1F2		37	34,36	14,15,16	35				
P64	1F3		37	34,36	14,15,16	17				
P65	2F0		30	32,12	11,33,7	29				
P66	2F1			32,12	11,33,7	13				
P67	2F2			28,8	6,9,7	5				
P68	2F3			28,8	6,9,7	27				
P69	3F0		25	22,2	3,23,4	21				
P70	3F1		25	22,2	3,23,4	24				
P71	$4\overline{1}$	2	26	Open	Open	31	2	19		

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Wave-form	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1F0 Propagation delay time (t_{PLH})	1	19	16,37	39,20	18	7	26	ns	14,15
R2	1F1		38		39,20	40				
R3	1F2		14		34,36	35				
R4	1F3		15		34,36	17				
R5	2F0		11	7,30	32,12	29				
R6	2F1		33		32,12	13				
R7	2F2		6		28,8	5				
R8	2F3		9		28,8	27				
R9	3F0		3	4,25	22,2	21				
R10	3F1		23	4,25	22,2	24				
R11	1F0		39	37	19,38,16	18				
R12	1F1		20		19,38,16	40				
R13	1F2		34		14,15,16	35				
R14	1F3		36		14,15,16	17				
R15	2F0		32	30	7,11,33	29				
R16	2F1		12		7,11,33	13				
R17	2F2		28		7,6,9	5				
R18	2F3		8		6,9	27				
R19	3F0		22	25	3,23,4	21				
R20	3F1		2	25	3,23,4	24				
R21	1F0	2	39	16	19,38,37	18	10	35		
R22	1F1		20		19,38,37	40				
R23	1F2		34		14,15,37	35				
R24	1F3		36		14,15,37	17				

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Symbol	Parameter	Waveform	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
	2F0 Propagation delay time (t_{PLH})	2	32	7	11,33,30	29	10	35	ns	14,15
	2F1		12		11,33,30	13				
	2F2		28		6,9,30	5				
	2F3		8		6,9,30	27				
	3F0		22	4	3,23,25	21				
	3F1		2		3,23,25	24				
	1F0		16	37,39,20	19,38	18				
	1F1					40				
	1F2				37,34,36	14,15	35			
R34	1F3					17				
R35	2F0		7	30,32,12	11,33	29				
R36	2F1				30,32,12	11,33	13			
R37	2F2				30,28,8	6,9	5			
R38	2F3					6,9	27			
	3F0		4	25,22,2	3,23	21				
	3F1					24				
R41	1F0	1	16	Open	39,19,20, 38,37	18		42		
	1F1					40				
	1F2				14,34,15, 36,37	35				
R44	1F3					17				
	2F0		7		11,32,33 12,30	29				

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* TABLE XV Propagation delay time (t_{PHL} and t_{PLH}) test - Continued

Test condition	Parameter	Waveform	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R46	2F1 Propagation delay time (t_{PLH})	1	7	Open	11,32,33, 12,30	13	10	42	ns	4 15
R4	2F2				6,28,9, 8,30	5				
R48	2F3				6,28,9, 8,30	27				
R49	3F0		4		3,22,23, 2,25	21				
R50	3F1		4		3,22,23, 2,25	24				
R51	1F0	2	37	16	19,39,38, 20	18		35		
R52	1F1				19,39,38, 20	40				
R53	1F2				14,34,15, 36	35				
R54	1F3				14,34,15, 36	17				
R55	2F0		30	7	11,32,33, 12	29				
R56	2F1				11,32,33, 12	13				
R57	2F2				6,28,9,8,	5				
R58	2F3				6,28,9,8,	27				
R59	3F0		25	4	3,22,23, 2	21				
R60	3F1		25	4	3,22,23, 2	24				
R61	1F0	1	37	39,20	19,38,16	18		42		

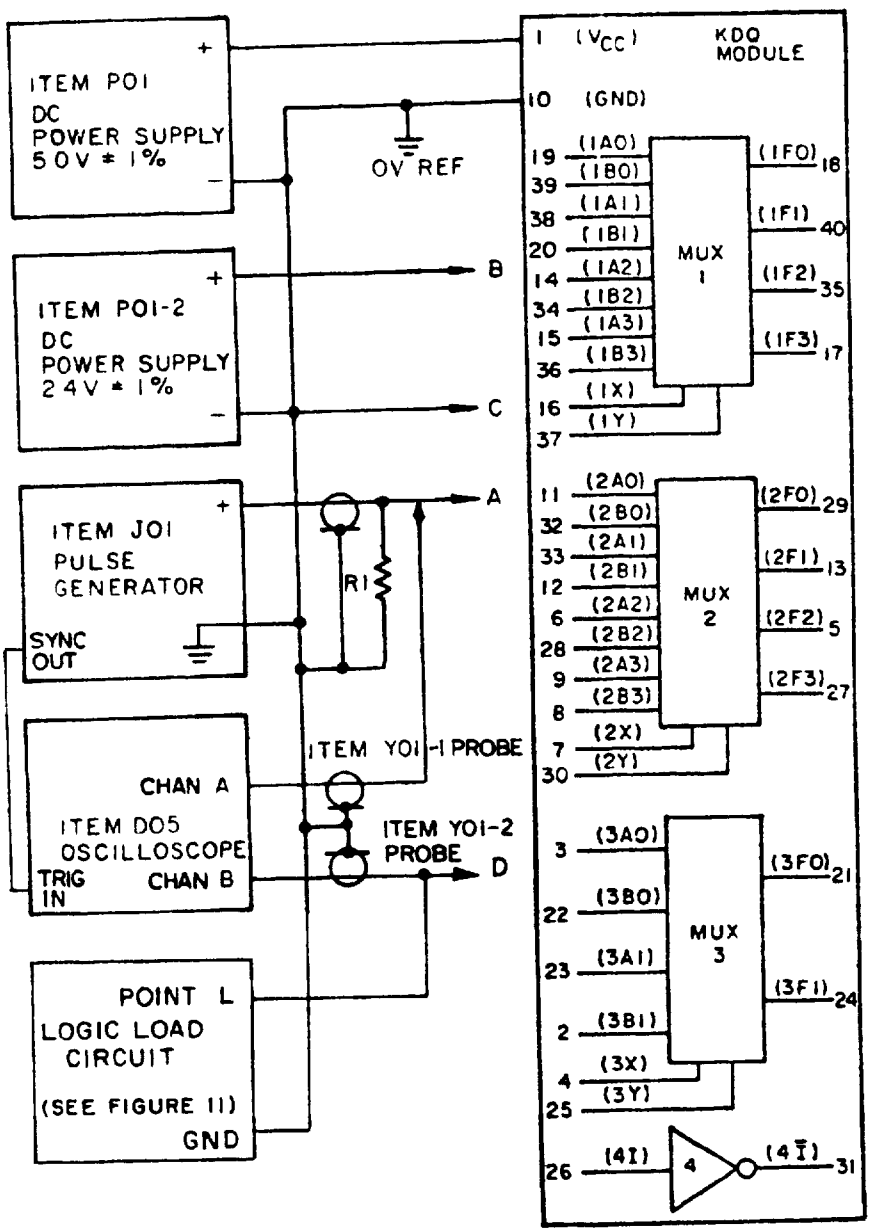
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* TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Wave-form	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
962	1F1 Propagation delay time (t_{p_H})	1	37	39,20	19,38,16	40	10	42	ns	14,15
963	1F2		37	34,36	14,15,16	35				
9[illegible]	1F[illegible]		[illegible]7	34,36	11,15,16	17				
[illegible]	2F0		30	32,12	11 33,7	29				
[illegible]66	2F1			32,12	11,33,7	13				
[illegible]67	2F2			28,8	6,9,7	5				
968	2F3			28,8	6,9,7	27				
969	3F0		25	22,2	3,23,4	21				
R70	3F1		25	22,2	3,23,4	24				
R71	4T	2	26	Open	Open	31	4	28		

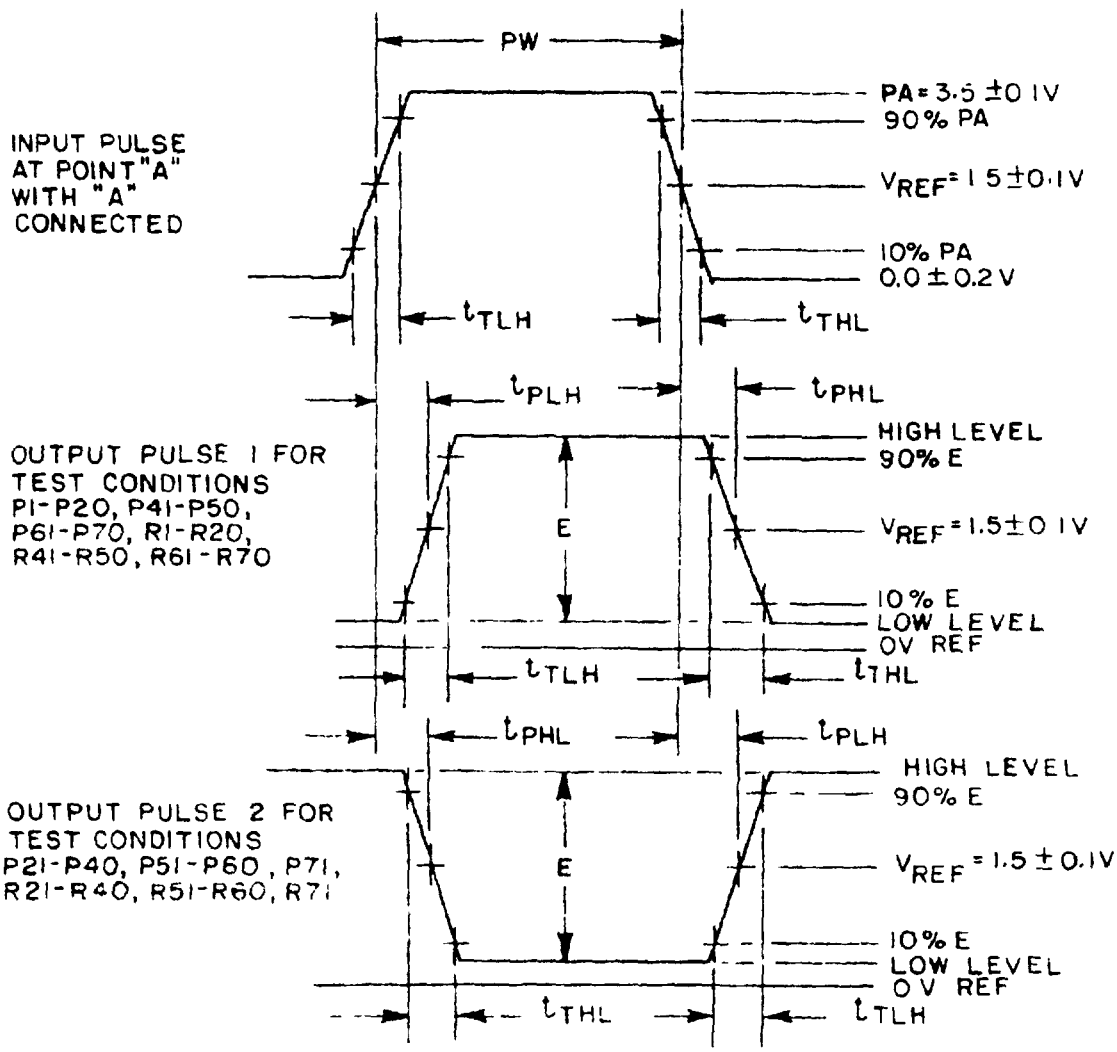
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NOTE R1 51Ω ±5%, 1 W.

* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIV and XV.

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NOTES
1. Input pulse parameters
   PW = 200 ns ±10%
   t_{TLH} = t_{THL} = 10 ±2 ns
   PRR = 1 kHz ±10
   Before making these measurements verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V

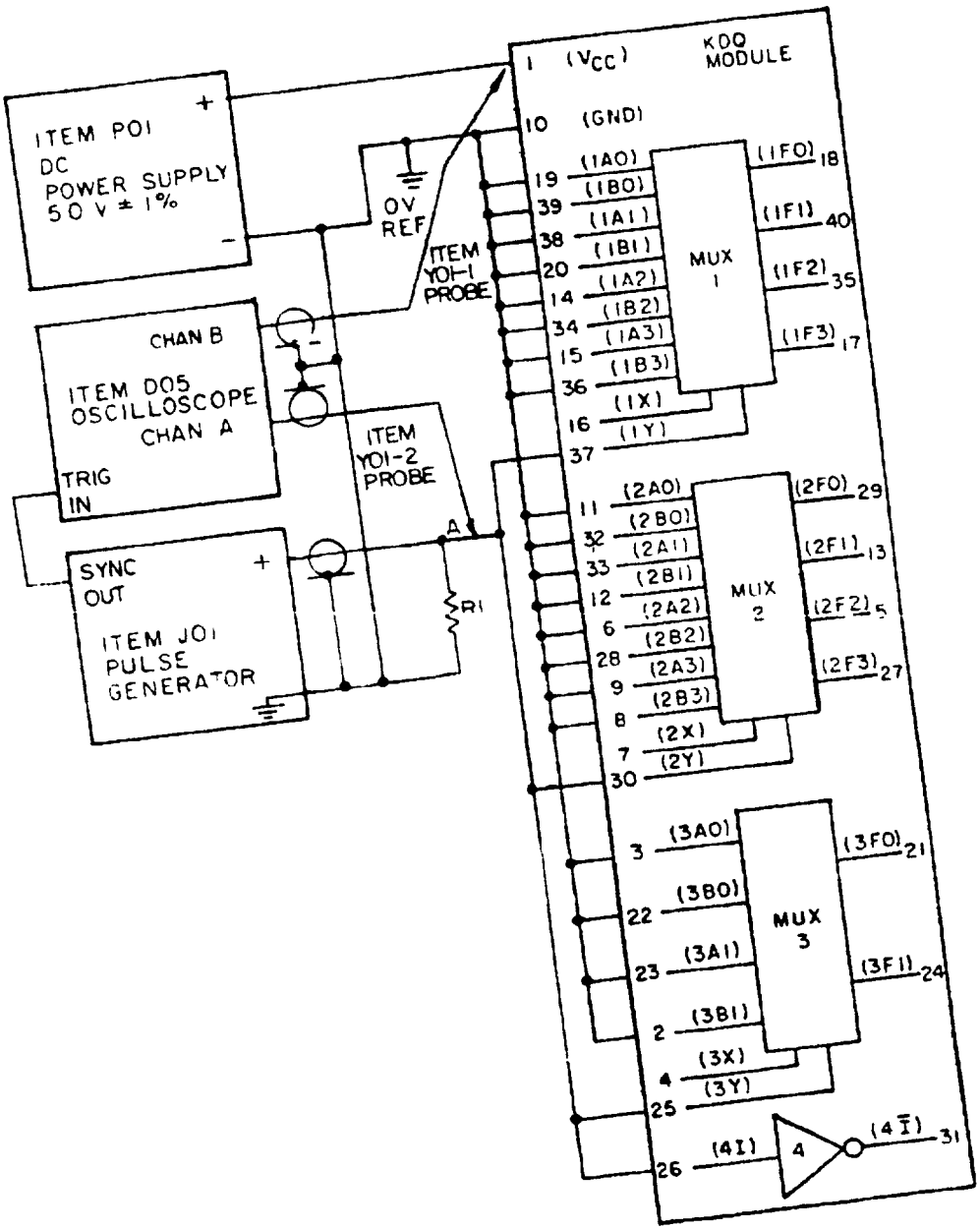
★ FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables VIV and XV

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TABLE XVI Power supply transient amplitude test

Test condition	Parameter	Initial limits 25°±5°C ambient		unit	Test figure
		Min	Max		
3	Power supply transient amplitude	0	400	mV	16,17

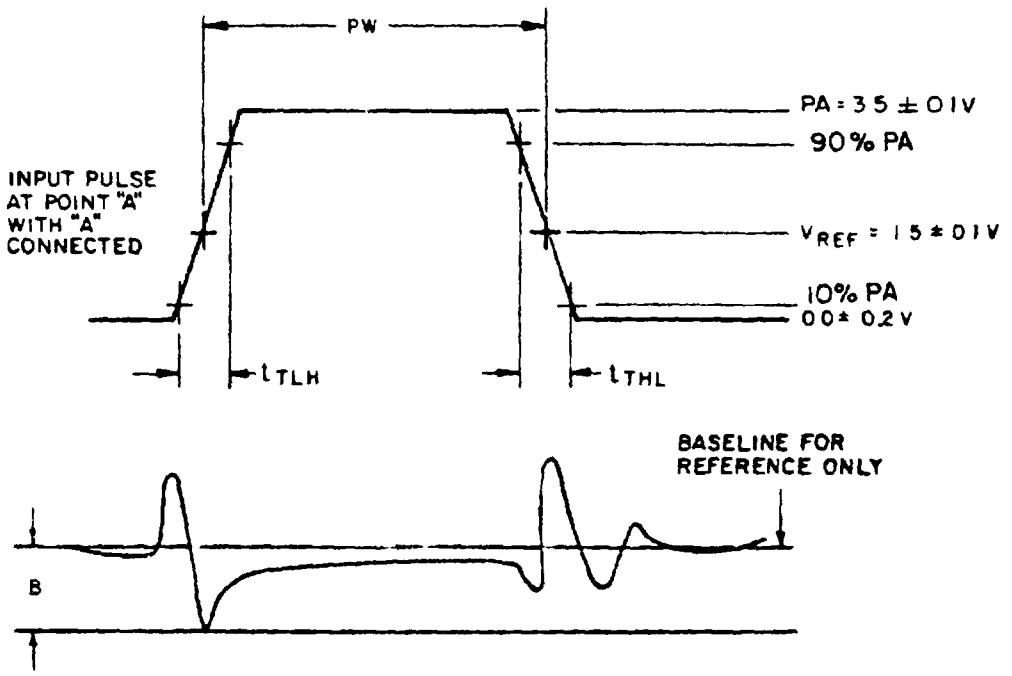
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NOTE R1 - 51Ω ±5%, 1 W.

* FIGURE 16 Power supply transient amplitude test circuit for table XVI

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NOTES

1. Input pulse parameters
   PW = 200 ns ±10%
   $t_{TLH} = t_{THL}$ = 10 ±1 ns
   PRR = 1 MHz ±10%
2. B = measurement from the baseline to the most negative excursion of the waveform

* FIGURE 17. Power supply transient amplitude test circuit waveforms for table XVI

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TABLE XVII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
T1	1F0 Output crosstalk, low level	25,26, 30,37	39	19,20, 38 / 14,15, 34,36 / 11,12, 32,33 / 6,8, 9,28 / 2,3, 22,23	18	0	800	mV	18,19
T2	1F1		20	19,38, 39 / 14,15, 34,36	40				
T3	1F2		34	19,20, 38,39 / 14,15, 36	35				
T4	1F3		36	14,15, 34	17				
T5	2F0		32	14,15, 34,36 / 11,12, 33	20				
T6	2F1		12	11,32, 33	13				
T7	2F2		28	11,12, 32,33 / 6,8,9	5				
T8	2F3		8	6,9, 28	27				
T9	3F0		22	6,8, 9,28 / 2,3, 23	21				
T1	3F1		2	6,8, 9,28 / 3,22, 23	24				
T1	4T	25,30, 37	26	6,8, 9,28 / 2,3, 22,23	31				

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TABLE XVII Output crosstalk (low-level) test - Continued.

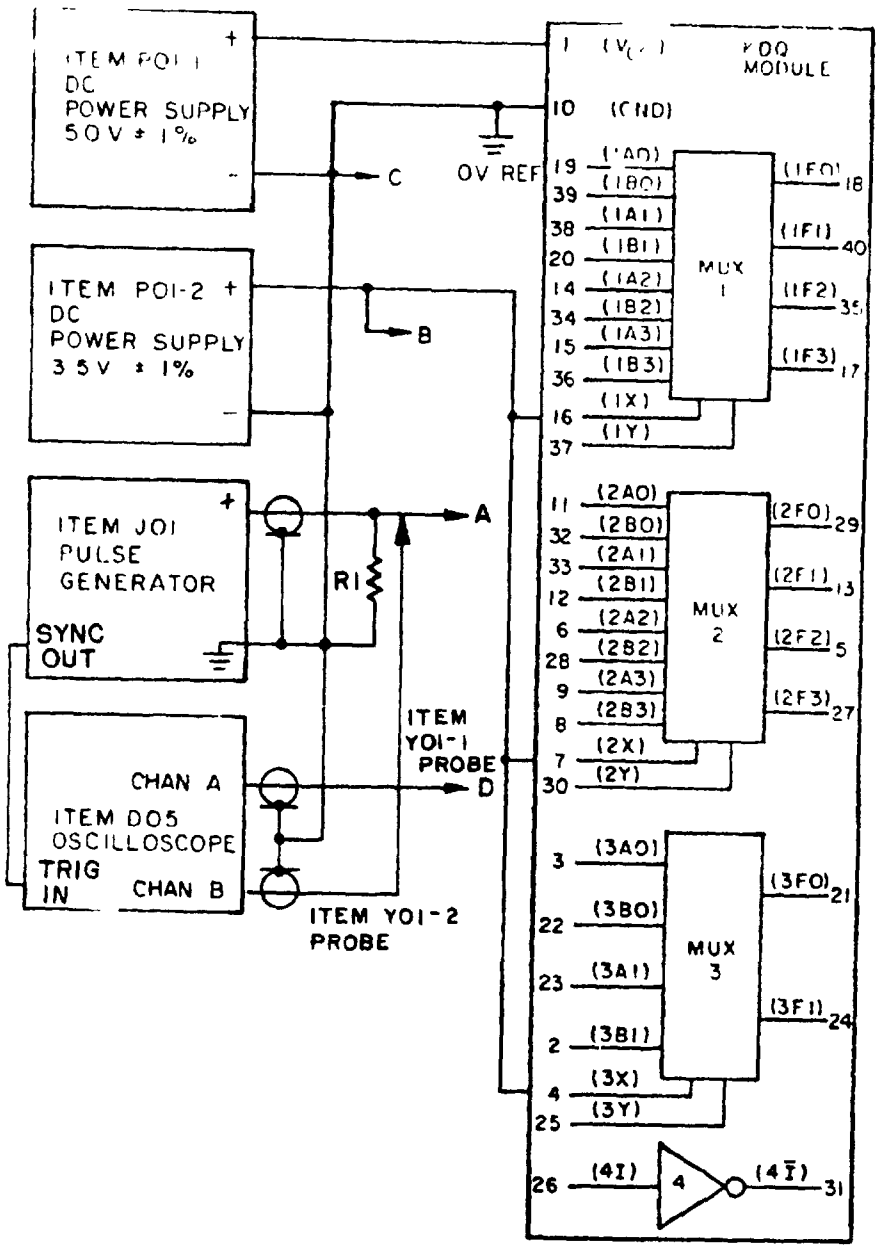
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T12	1F0 Output crosstalk, low level	25,26, 30,37	20,34, 36,39	2,3,6,8,9,11,12, 14,15,19,22,23, 28,32,33,38	18	0	400	mV	18,19
T13	1F1				40				
T14	1F2				35				
T15	1F3				17				
T16	2F0		8,12, 28,32	2,3,6,9,11,14,15, 19,20,22,23,33, 34,36,38,39	29		450		
T17	2F1				13				
T18	2F2				5				
T19	2F3				27				
T20	3F0		2,22	3,6,8,9,11,12,14, 15,19,20,23,28, 32,33,34,36,38,39	21				
T21	3F1		2,22	3,6,8,9,11,12,14, 15,19,20,23,28, 32,33,34,36,38,39	24				

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TABLE XVIII Output crosstalk (high-level) test.

Test condition	Parameter	Connect							Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin				D to pin	Min	Max		
U1	1F0 Output crosstalk, high level	20,25,26, 30,34,36	14,15, 19,38	37,39	11,12, 32,33	6,8, 9,28	2,3, 22,23	18	0	400	mV	18,19
U2	1F1	39,25,26, 30,34,36		20,37				40				
U3	1F2	20,39,36, 30,25,26		34,37				35				
U4	1F3	20,39,34, 30,25,26		36,37				17				
U5	2F0	37,12,8, 28,25,26	6,9, 11,33	14,15, 34,36	32,30	19,20, 38,39		29				
U6	2F1	37,32,8, 28,25,26			12,30			13				
U7	2F2	37,12,32, 8,25,26			28,30			5				
U8	2F3	37,12,32, 28,25,26			8,30			27				
U9	3F0	37,30,2, 26	3,23		6,8,9, 11,12, 28,32, 33		22,25	21				
U10	3F1	37,30, 22,26	3,23		6,8,9, 11,12, 28,32, 33		2,25	24				
U11	4Ī	37,30, 25	Open		6,8,9, 11,12, 28,32, 33		2,3,22, 23,26	31				

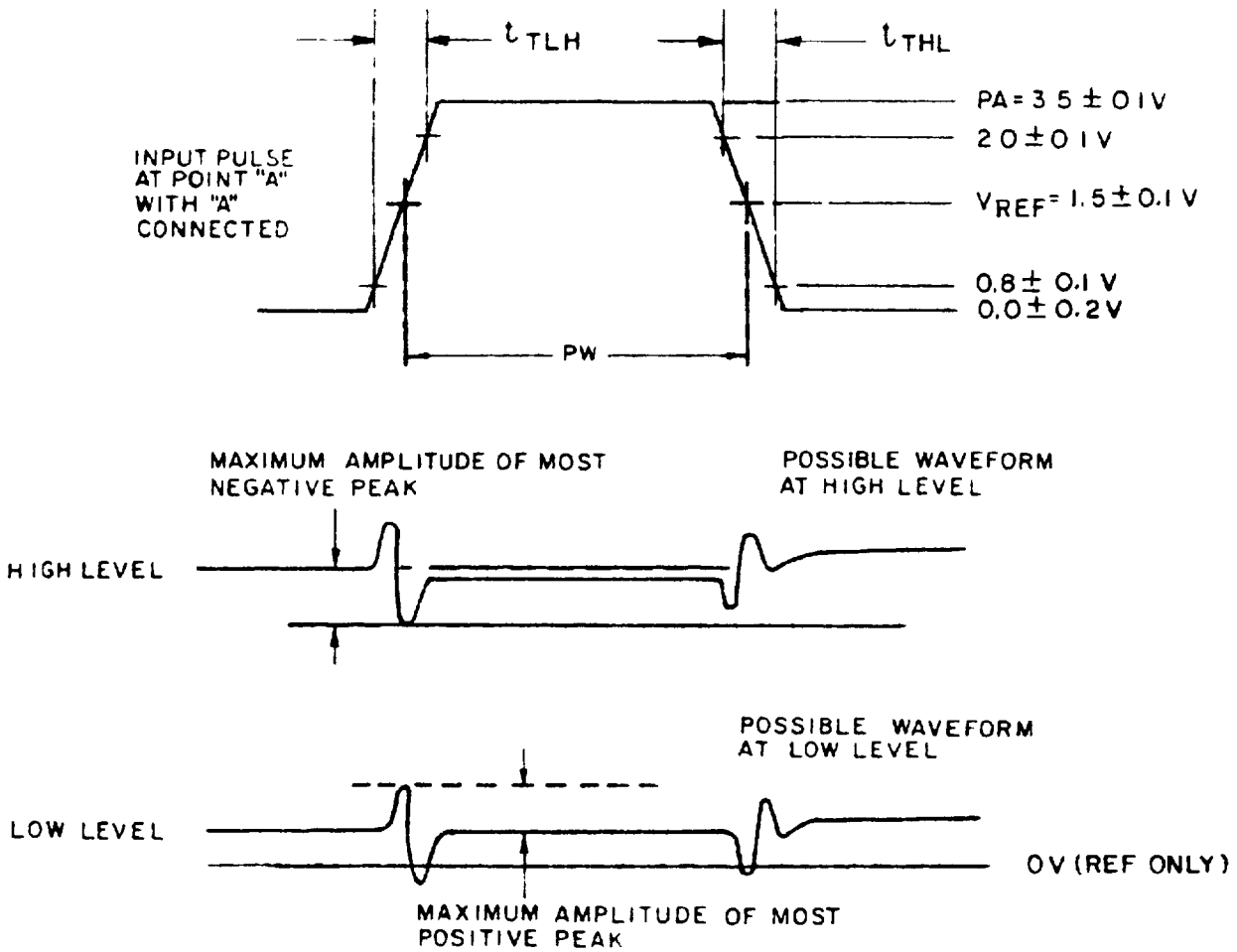
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NOTES
1. R1 = 51Ω ±5 , 1 W
2. Load each output as specified on figure 11.

* FIGURE 18. Out...t crosstalk (low-level and high-level) test circuit for tables XVII and XVIII.

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NOTE Input pulse parameters
PW = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1 mHz ±10%.

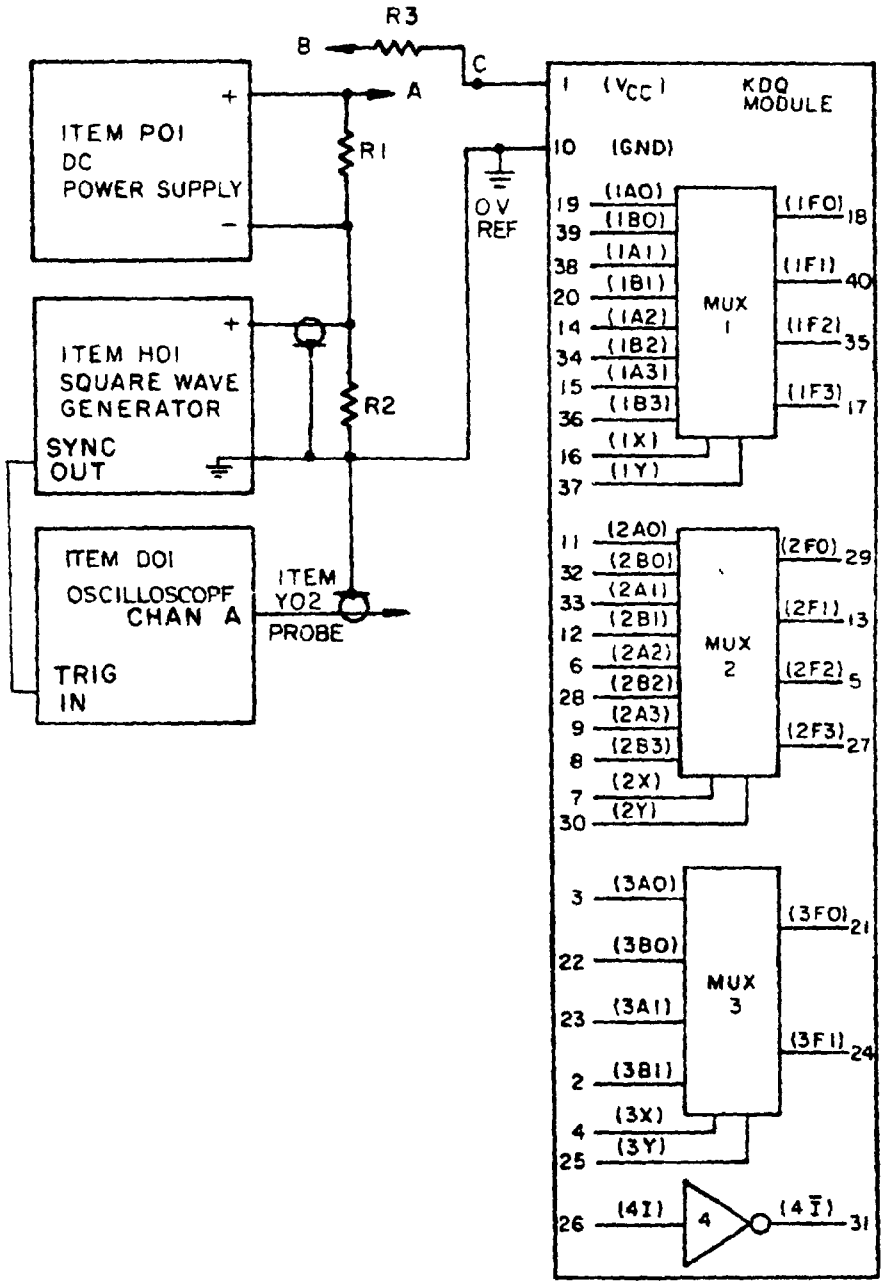
★ FIGURE 14. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVII and XVIII.

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TABLE XIX Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient rise time	1	2	ms	20,21

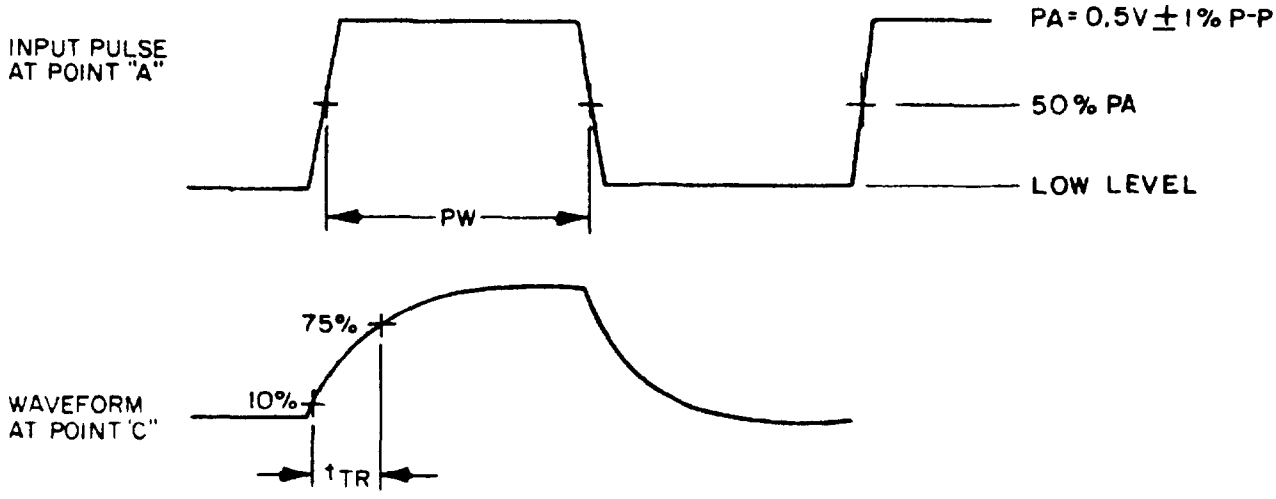
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NOTES
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1 4 W

★ FIGURE 20. Power supply transient rise time test circuit for table XIX.

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NOTES
1. Input pulse parameters
   PW = 10 ms ±10%.
   PRR = 50 Hz (reference).
   Rise and fall time of input < 100 ns (reference).
2. With point "A" open, adjust input as shown.
3. With point "A" open, adjust item P01 such that the low level of the input is at 0 ±0.01 V.
4. Connect "A" to "B", recheck the low level and pulse parameters at point "A".
5. Measure transient rise time (t_{TR}) at point "C".

* FIGURE 21. Power supply transient rise time test circuit waveforms for table XIX.

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